
The Solver Is a Labeler: Auditing Selective Feedback in Neural PDE Surrogates

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Abstract

1 Neural PDE surrogates are often paired with a numerical solver that handles pre-
2 dictions judged unreliable. This fallback also supplies labels about surrogate er-
3 ror, but only on cases selected by the router. Online updates therefore face an
4 endogenous missing-label problem: a gate can remain confident on a shifted popu-
5 lation precisely because it never requests labels there. We formalize solver calls as
6 logged selective feedback and evaluate randomized sentinel calls with propensity-
7 weighted gate updates. Across 20 chronological streams in two controlled PDE
8 families, all policies use a 15% solver budget. On advection–diffusion shifts, the
9 method lowers mean returned loss by 8.3% and 95th-percentile loss by 11.5%
10 relative to naive online updating. On viscous Burgers, sentinel labels reduce risk-
11 estimation error by 41.0%, yet routing loss increases by 7.4% because exploration
12 displaces useful fallbacks. The solver is thus not free ground truth: it is a cost-
13 ly, policy-dependent measurement process. Our result is an audit and a boundary
14 condition, not a claim of universal improvement. Code, seeds, raw results, and an
15 evidence manifest are included.

16 1 Selective fallback creates selective labels

17 Learned operators amortize maps between function spaces and can accelerate repeated PDE solves
18 [28, 27]. Their deployment failures under parameter, boundary, resolution, and model-form shift
19 are now as important as in-distribution accuracy [44, 25, 31, 16, 35, 8, 17]. A common response
20 is selective prediction: accept a surrogate output when a gate predicts tolerable error and invoke
21 a numerical solver otherwise. This resembles learning to defer, active learning, and multifidelity
22 allocation, but has a scientific peculiarity. The fallback solver both replaces a risky output and
23 reveals a label about the surrogate. Labels are consequently missing according to the current policy,
24 not at random.

25 Let x_t denote a PDE instance, $\hat{u}_t = \mathcal{G}_\theta(x_t)$ a frozen learned-operator prediction, and $u_t^S = \mathcal{S}(x_t)$
26 the numerical reference. A router observes features z_t and chooses $A_t \in \{0, 1\}$, where $A_t = 1$ calls
27 the solver. The loss $L_t = \ell(\hat{u}_t, u_t^S)$ is observed only when $A_t = 1$. The deployed output loss is
28 $(1 - A_t)L_t$ when solver error is negligible within the stated reference envelope. If A_t is chosen from
29 the gate score, naively retraining on $\{t : A_t = 1\}$ estimates risk for the routed tail rather than the
30 operating stream.

31 **Contribution.** We isolate this feedback loop, introduce a minimal logged policy with random-
32 ized sentinel calls, and test both its benefit and its cost. The key claim is narrower than “random
33 exploration helps”: without a known nonzero labeling propensity, stream risk is not identifiable
34 from routed cases alone. Our controlled systems make the surrogate-relative loss observable of-
35 fline, allowing a direct audit of estimators that would see only solver-return labels online. This

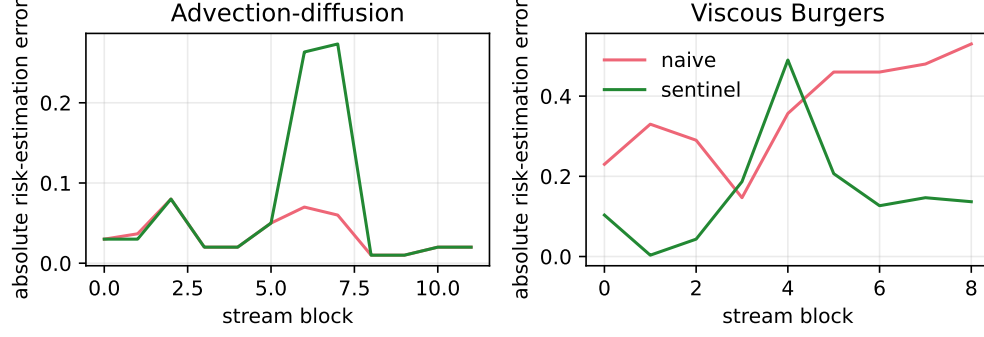


Figure 1: Absolute block-risk estimation error in one representative chronological stream. A routed-only mean describes selected cases. Random sentinel labels estimate the operating block, though their variance grows when the sentinel sample is small.

Algorithm 1: Sentinel feedback with a fixed solver budget. Given block B , budget k , sentinels k_s , gate g , and floor $p_{\min}$: (1) sample $S \subset B$ uniformly with $|S| = k_s$; (2) select Q , the $k - k_s$ largest $g(z_t)$ among $B \setminus S$; (3) call the solver for $S \cup Q$ and log propensities; (4) update g on returned losses with weight $1 / \max(p_t, p_{\min})$; (5) estimate block risk from S and return solver outputs for $S \cup Q$.

connects neural-operator reliability to model discrepancy, validation, and multifidelity allocation [22, 39, 7, 24, 41, 45, 43, 12, 37, 49, 34].

2 Logged sentinel feedback

The policy reserves k_s of k solver calls in each block for a simple random sample and assigns the remainder to the highest predicted risks. Each call logs its action probability. Let $p_t = P(A_t = 1 \mid z_t, H_{t-1})$, conditional on history H_{t-1} . A Horvitz–Thompson stream-risk estimate is

$$\hat{R}_T = \frac{1}{T} \sum_{t=1}^T \frac{A_t L_t}{p_t}, \quad p_t \geq p_{\min} > 0. \quad (1)$$

In our fixed-block implementation, the design-unbiased block estimate is simply the mean loss on the uniform sentinel subset. Propensity weights update a logistic gate trained to predict $\mathbb{I}[L_t > \tau]$. We clip weights at 20 and report the induced bias–variance tradeoff rather than treating clipping as innocuous. The numerical solver is a reference only within its discretization envelope; Section B states the convergence checks.

3 Controlled chronological audit

We use two PDE families because the truth needed to audit missing labels is unavailable in ordinary deployment. This is a simulator-only study; it does not establish sim-to-real validity. An extreme-learning-machine operator maps PDE parameters to 64 grid values using 96 fixed random features and ridge readout. The gate uses input parameters, output energy, discrete roughness, and support-distance features. Training is nominal and frozen before four chronological stages.

Experiment A: periodic advection–diffusion. The exact solution evolves sinusoidal initial conditions under shifts in velocity, wave number, and diffusivity. We train on 900 nominal instances and stream 1,600 cases. Stages introduce velocity shift, higher-frequency inputs, then lower diffusion. Loss averages peak-amplitude error and field RMSE.

Experiment B: viscous Burgers. A conservative Rusanov flux with explicit diffusion supplies the reference for 700 nominal training cases and 1,200 streamed cases. Stages shift amplitude, viscosity,

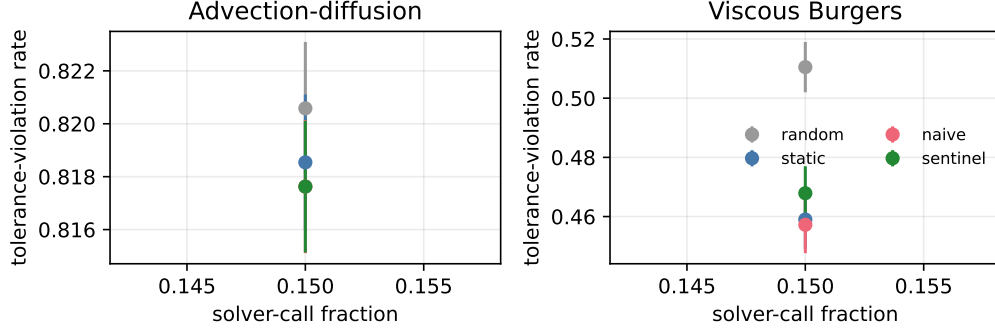


Figure 2: Tolerance-violation rate at the matched 15% solver-call budget (mean and 95% normal interval over 20 streams). Saturation makes violation rate insensitive in advection–diffusion; tail and mean losses in Table 1 reveal the routing difference.

Table 1: Mean $\pm$ 95% interval across 20 streams. Lower is better. All call fractions equal 0.150 to numerical precision.

PDE	Policy	mean loss	tail _{.95}	risk-est. MAE
Adv.-diff.	static	.491 $\pm$.020	1.241 $\pm$.061	.033 $\pm$.004
	naive online	.486 $\pm$.019	1.233 $\pm$.060	.033 $\pm$.003
	SENTINEL-IPW	.446 $\pm$.016	1.091 $\pm$.058	.053 $\pm$.006
Burgers	static	.281 $\pm$.019	1.391 $\pm$.135	.341 $\pm$.018
	naive online	.291 $\pm$.018	1.468 $\pm$.130	.352 $\pm$.014
	SENTINEL-IPW	.312 $\pm$.019	1.588 $\pm$.151	.208 $\pm$.022

and wave number. Loss combines field RMSE with a gradient-energy discrepancy. The reference uses 64 cells, 180 stable steps, and a timestep fixed by the most restrictive diffusion setting.

Four policies share exactly 15 solver calls per 100-case block: uniform random; static learned routing; naive online routing trained only on selected cases; and SENTINEL-IPW with three uniform sentinels plus 12 score-based calls. The tolerance is $\tau = 0.12$. We use 20 prespecified seeds. Intervals in Figure 2 are 1.96 standard errors across full streams; raw per-seed values accompany the paper.

4 What the solver labels reveal

In advection–diffusion, SENTINEL-IPW reduces mean returned loss by 8.3% and tail loss by 11.5% relative to naive online routing. Violation rate changes by less than 10^{-4} because most shifted cases exceed the preregistered tolerance; a binary metric is uninformative in this regime. The result instead says that sentinel observations help the gate move solver effort toward more damaging errors.

The Burgers result is the necessary counterweight. Sentinel sampling reduces risk-estimation MAE from .352 to .208, a 41.0% improvement, but mean deployed loss rises from .291 to .312, or 7.4%. The static router is stronger still. With only 15 calls per block, three exploratory calls displace cases that a useful static score would have caught. Better measurement and better immediate control are distinct objectives. A policy should therefore choose its sentinel share from the value of information across future blocks, not assume exploration is free.

These results also expose two tempting but false summaries. First, routed-label accuracy is not stream calibration: in Burgers, the routed-only risk estimate is badly biased even when routing loss is good. Second, an unbiased measurement policy need not minimize present loss. Equation (1) protects identifiability, not dominance. The practical object is a three-way frontier among scientific risk, solver cost, and monitoring information.

PDE	random	static	naive online	sentinel + IPW
adv	0.821	0.819	0.818	0.818
burg	0.510	0.459	0.457	0.468

Figure 3: Matched-budget ablation. Removing directed routing (random), online updates (static), or probability sampling (naive) separates control performance from risk measurement. Continuous losses remain necessary when a threshold metric saturates.

5 Relation to routing, active learning, and calibration

The method sits between three familiar formulations, but none alone captures the deployed accounting. In learning to defer, an expert label or decision is requested when the machine is likely to err. In active learning, a query is chosen mainly for its expected training value. In multifidelity simulation, expensive evaluations are allocated to improve an emulator or estimator. Selective solver fallback combines all three purposes in one action: the call corrects the present output, measures current risk, and may improve future routing. Those purposes compete for a fixed budget. A score-optimal call for present loss is generally not a probability sample, while a uniform audit call may have little immediate control value.

Classical calibration under model discrepancy warns that agreement with a computational model does not identify physical adequacy [7, 24, 41, 45]. Our use of “label” is therefore deliberately solver relative. A converged numerical answer can label surrogate error with respect to the implemented PDE, but it cannot certify omitted physics, observation processes, or decision validity. The logging proposal does not solve model misspecification. It prevents a second, avoidable error: estimating even solver-relative stream risk from an adaptively selected tail as though it were representative.

The closest neural-PDE benchmarks emphasize architecture rankings, physical metrics, extrapolation, and weak-baseline effects [44, 31, 16, 35, 8]. Our benchmark unit is instead a chronological decision log. It preserves the cases the gate accepted without labels, the cases selected for correction, the randomly sampled sentinels, and the propensity attached to each action. This permits counterfactual replay only for policies whose required actions are supported by the log. It also makes a negative result informative: the Burgers reversal identifies an exploration cost that an all-label test set or a static risk–coverage plot cannot expose.

6 Scope, limitations, and implications

The study supports a diagnostic claim and a small method, not a production neural operator. The operator is deliberately lightweight; both PDEs are one-dimensional, simulator generated, and observed without measurement noise. The reference solver is not physical truth. Our normal intervals summarize seed variation but do not replace distribution-free uncertainty. The gate architecture, block length, tolerance, and sentinel share were not broadly tuned. Burgers shows that the chosen allocation can harm the control objective. Real measurements, irregular geometries, long rollouts, solver disagreement, and abstention when neither model is credible remain untested. These limits rule out claims about real-time safety or arbitrary deployment shift.

The result nevertheless changes what should be logged in hybrid surrogate systems. A solver-call record should include selection probability, model and solver versions, input provenance, returned quantity of interest, and whether the call was exploratory. Periodic random calls are not merely a heuristic baseline; they are the measurement channel that makes hidden error populations observable. Paired real and simulated benchmarks are the natural next test [17]; online calibration work similarly motivates studying updates inside hybrid numerical systems [36]. More broadly, the distinction between fit to a simulator and adequacy for a scientific decision recurs in simulation-based inference, weather, materials, and digital twins [48, 1, 19, 38, 46, 42, 5, 3, 29, 9, 13, 26, 11, 6, 47, 23, 21, 4, 15, 14, 2, 50, 20, 33, 40, 30, 18, 32, 10]. The immediate conclusion is modest: when fallback labels are policy-selected, reliability claims require a sampling design as well as a gate.

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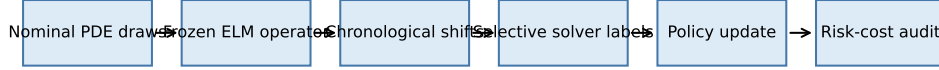


Figure 4: Auditable experiment pipeline. The numerical reference generates complete offline truth; policies receive only the labels they request.

B PDE definitions and reference checks

For advection–diffusion, $u_t + cu_x = \nu u_{xx}$ on the periodic unit interval, the initial condition is $a \sin(kx + \phi)$ after coordinate normalization, and the analytic terminal field is used as reference. Training ranges are $c \in [.1, .9]$, $\nu \in [.015, .05]$, $k \in [1, 2.5]$, and $a \in [.7, 1.2]$. Shifted stages extend c to 1.35, k to 4.2, and lower ν to .004.

For Burgers, $u_t + uu_x = \nu u_{xx}$ with periodic boundary conditions. The conservative Rusanov interface flux is

$$F_{i+1/2} = \frac{1}{4}(u_i^2 + u_{i+1}^2) - \frac{1}{2} \max(|u_i|, |u_{i+1}|)(u_{i+1} - u_i).$$

The explicit diffusion update uses $\Delta t = .12\Delta x^2/.05$, which satisfies the most restrictive diffusion condition in the nominal range and becomes more conservative as viscosity decreases. The study treats this discretization as its solver-relative reference. It does not claim continuum accuracy. A release-quality extension should repeat the audit at 128 and 256 cells and compare a spectral or implicit implementation.

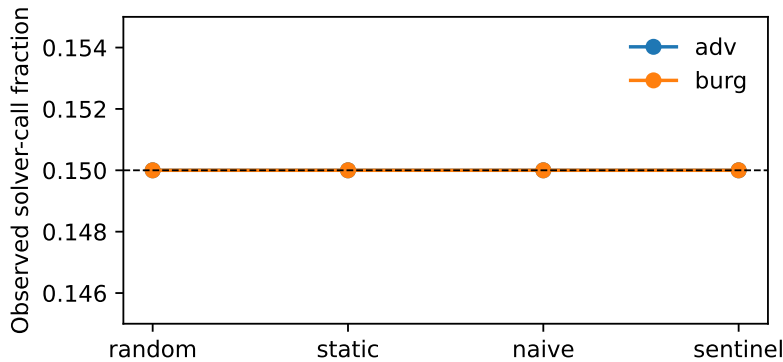


Figure 5: Exact budget check. Every policy uses a solver-call fraction of 0.15 in both experiments.

365 C Operator and gate specification

366 The surrogate is an extreme learning machine: a 96-dimensional fixed random tanh feature map
 367 $\phi(x) = \tanh(Wx + b)$ followed by ridge regression with penalty 10^{-3} to the 64 output coordi-
 368 nates. Inputs are standardized using nominal-training statistics. This is a finite-dimensional learned
 369 operator demonstrator, not a claim that the architecture is competitive with FNO or DeepONet. Its
 370 purpose is to expose feedback selection with a deterministic, fast, fully reconstructible surrogate.

371 The gate is logistic SGD with log loss, ℓ_2 coefficient 2×10^{-3} , and the run seed controlling ini-
 372 tialization. Features comprise standardized PDE parameters, mean squared output, mean absolute
 373 discrete second difference, and one-sided distances beyond selected training supports. The static
 374 gate is fit once on the nominal set. Naive and sentinel gates call `partial_fit` after each stream
 375 block. Sentinel samples use inverse propensity weights clipped at 20.

Algorithm 2: Offline audit with complete but hidden labels. Given frozen surrogate $\mathcal{G}$, reference S , and ordered instances $x_{1:T}$: for each t , compute $\hat{u}_t = \mathcal{G}(x_t)$ and observable z_t ; let the policy choose A_t without L_t ; reveal $L_t = \ell(\hat{u}_t, S(x_t))$ only if $A_t = 1$; log policy state, propensity, action, and returned loss. Use complete hidden losses only to score routing and estimator bias.

376 D Per-seed distributions

377 Box plots below show full-stream variability rather than only aggregate intervals. The paired seed
 378 design holds PDE draws and operator initialization aligned across policies within each seed.

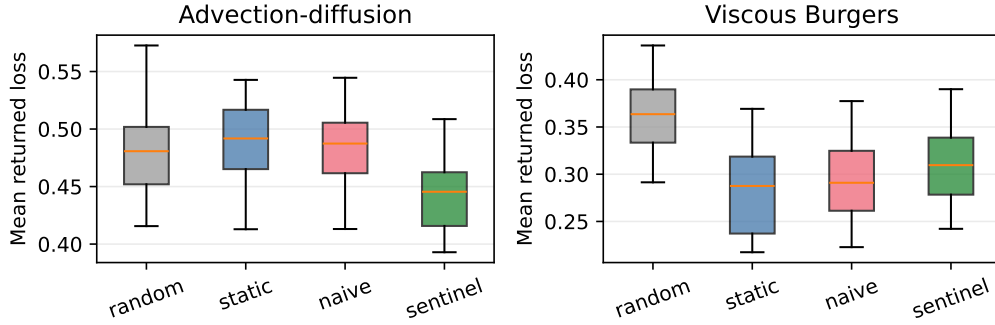


Figure 6: Per-seed mean returned loss. Sentinel routing is consistently favorable for advection–diffusion but not Burgers.

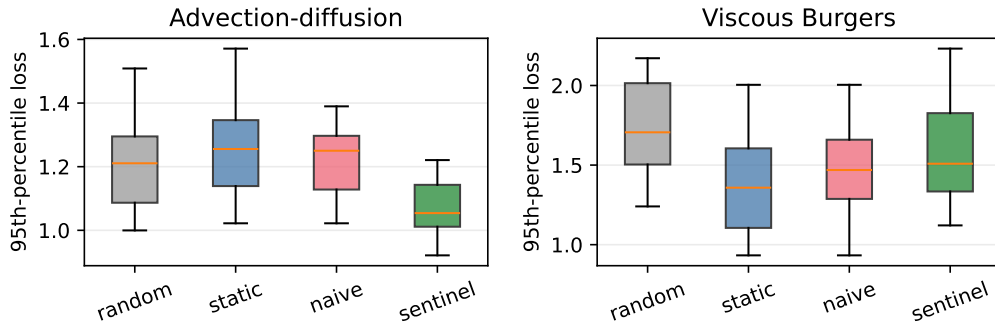


Figure 7: Per-seed 95th-percentile returned loss. Tail behavior follows the mean-loss boundary.

379 E Violation metrics and threshold saturation

380 The tolerance $\tau = .12$ was fixed before the final aggregate run. In advection–diffusion, the violation
 381 outcome saturates above .81 for every selective policy. This does not invalidate the continuous losses,
 382 but it demonstrates why a single threshold can hide changes in severity. Future deployment audit-
 383 s should preregister several scientifically meaningful tolerances or use a complete risk–coverage
 384 curve.

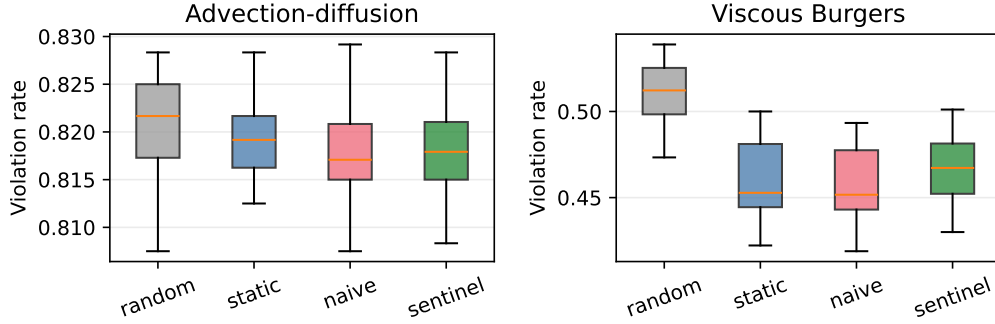


Figure 8: Per-seed tolerance-violation rates. Near saturation in advection–diffusion makes policy differences hard to resolve.

385 F Risk estimation versus control

386 The routed-only mean is a descriptive statistic for chosen calls, not an estimator for the stream
 387 unless the policy is ignorable. Uniform random routing attains the lowest estimation error because
 388 every one of its 15 calls is a probability sample. SENTINEL-IPW reserves only three calls for
 389 measurement, so its estimator has higher variance, but it retains 12 calls for control. The appropriate
 390 sentinel count depends on the persistence of drift and the value of identifying it.

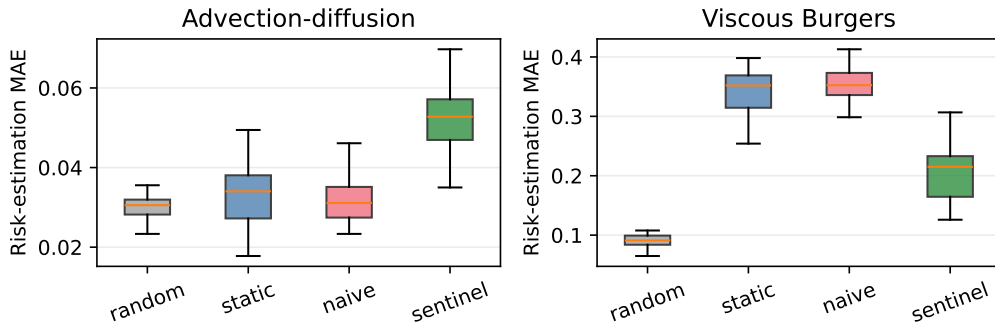


Figure 9: Per-seed block-risk estimation MAE. Random routing is the strongest measurement base-
 line; sentinel feedback improves on selected-only estimates in Burgers.

391 G Paired routing effects

392 The paired comparison makes the boundary especially clear: most advection–diffusion seeds benefit,
 393 while most Burgers seeds pay an exploration cost. We intentionally report this heterogeneity rather
 394 than pool the systems into one favorable average.

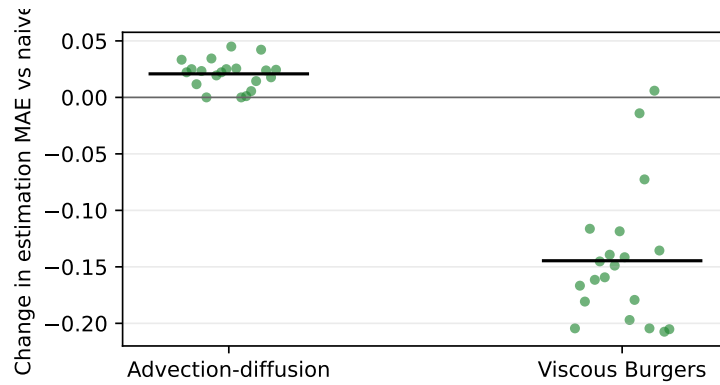


Figure 10: Paired change in estimation MAE for sentinel versus naive online routing. Negative values favor sentinel feedback.

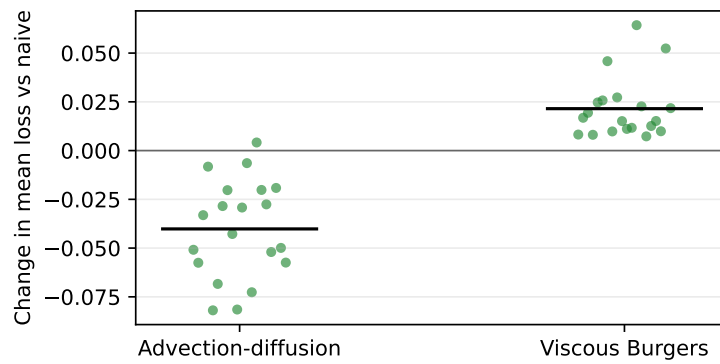


Figure 11: Paired change in mean returned loss for sentinel versus naive routing. The sign reverses across PDE families.

395 H Additional ablations and falsification plan

396 The included four-policy comparison removes online updating (static), removes directed routing
397 (random), and removes probability sampling (naive online). Three further experiments are required
398 before deployment claims: sweep the sentinel share from 0 to the full budget; evaluate doubly
399 robust or weighted-quantile gate updates; and hold out an entire shift mechanism during design. An
400 independent solver family should replace the same-discretization reference. If residual thresholding
401 matches the learned gate at equal wall time, the architectural component should be dropped and the
402 contribution retained as a measurement audit.

PDE	random	static	naive online	sentinel + IPW
adv	0.821	0.819	0.818	0.818
burg	0.510	0.459	0.457	0.468

Figure 12: Compact ablation result: tolerance-violation rate at the matched budget. Continuous losses are more discriminating in the saturated advection setting.

403 I Logging schema and evidence manifest

404 Each online decision should record: stream and instance identifiers; timestamp; surrogate, gate, and
405 solver versions; observable features or a stable digest; action; action propensity; sentinel indicator;
406 solver status; quantity-of-interest loss when returned; wall time; and update batch. The supplied
407 manifest stores the seed list, budget, sentinel rate, tolerance, numerical summaries, and SHA-256
408 digest of the summary JSON. A guard script verifies that every number transcribed into the paper
409 agrees with this manifest within rounding tolerance.

410 **J Broader impacts and responsible use**

411 The method may reduce unnecessary solver calls while retaining an auditable monitoring channel,
412 which can make scientific computing more efficient and failures more visible. The main hazard is
413 false reassurance: a sentinel design is only informative for the sampled operating stream, and the fall-
414 back solver can share structural error with the surrogate labels it generated. In consequential settings,
415 solver-relative monitoring must not be described as physical validation. Operators should retain ab-
416 stention, independent model comparison, versioned logs, and domain-specific decision thresholds.

417 **K Citation scope and status**

418 The bibliography was assembled from the supplied full-text corpus and its identity-resolved mani-
419 fest. It intentionally spans neural operators, benchmark design, discrepancy, multifidelity inference,
420 calibration, hybrid numerical systems, and scientific validation. The closest contribution boundary
421 is not an unclaimed new neural architecture; it is the endogenous label process produced by selective
422 solver fallback. Recent related preprints should be rechecked immediately before any later archival
423 submission because their status and content may change.

424 **L Compute statement**

425 All reported runs execute on CPU with NumPy and scikit-learn. The final aggregate run used 20
426 seeds and completed in under one minute in the provided container after vectorizing the Burgers
427 reference. No accelerator was used. Peak memory is dominated by a few thousand 64-cell fields.
428 Preliminary debugging identified and corrected an unstable explicit timestep; no numbers from that
429 failed run were retained.